OLS vs. MLE vs. GMM

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1 Overview

These notes[1] outline OLS, GMM, and MLE. Section 2 handles preliminaries. Section 3 introduces a running linear regression example, states classical assumptions, and identifies conditions under which the three methods coincide. Subsequent sections define OLS, MLE, and GMM and derive key conclusions (minimizers, unbiasedness, variances, and moment identities). The final sections discuss the relevance of the Gauss–Markov theorem and evaluate how point estimates and inference change when classical assumptions are relaxed.

2 Preliminaries

2.1 Estimators and their Properties

Let $W_T = (w_1, \ldots, w_T)$ denote the observed sample. An *estimator* of a k-dimensional parameter vector θ_0 is a measurable function $\hat{\theta}_T = \hat{\theta}(W_T)$ that maps data into $\mathbb{R}^k$.

Definition 1 (Bias and Unbiasedness)**.** The *bias* of $\hat{\theta}_T$ is $\mathbb{E}[\hat{\theta}_T] - \theta_0$ (when defined). The estimator is *unbiased* if $\mathbb{E}[\hat{\theta}_T] = \theta_0$.

Definition 2 (Consistency)**.** $\hat{\theta}_T$ is *consistent* for θ_0 if $\hat{\theta}_T \xrightarrow{p} \theta_0$ as $T \to \infty$, i.e.

$$\forall \epsilon > 0, \quad \Pr(\|\hat{\theta}_T - \theta_0\| > \epsilon) \to 0.$$

Definition 3 (Asymptotic Normality)**.** $\hat{\theta}_T$ is *asymptotically normal* if there exists a nonrandom positive definite matrix V such that

$$\sqrt{T}(\hat{\theta}_T - \theta_0) \xrightarrow{d} \mathcal{N}(0, V).$$

Definition 4 (Asymptotic Variance and Efficiency)**.** If $\sqrt{T}(\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}(0, V)$, then V is the *asymptotic variance* matrix. An estimator is *asymptotically efficient* (within a class) if it has the smallest asymptotic variance among estimators in that class under the maintained assumptions.

Remark 1 (Finite-Sample vs Asymptotic Statements)**.** Unbiasedness and exact finite-sample variance comparisons are finite-sample concepts. Consistency and asymptotic normality are large-sample concepts. In practice, econometric conclusions depend on *both* point estimation and inference (standard errors, confidence intervals, tests).

[1] Based on lecture by Prof. Fanyin Zheng (CBS).

2.2 The Cramér–Rao Lower Bound and Efficiency

Definition 5 (Score and Fisher information)**.** Let w have density $f(w \mid \theta)$ with $\theta \in \Theta \subset \mathbb{R}^k$. Define the *score* as

$$s(w, \theta) \equiv \nabla_\theta \log f(w \mid \theta),$$

and the *Fisher information matrix* as

$$I(\theta) \equiv \mathbb{E}\big[s(w,\theta)s(w,\theta)'\big] = -\mathbb{E}\big[\nabla^2_{\theta\theta} \log f(w \mid \theta)\big],$$

where the equality holds under standard regularity conditions (information matrix equality).

Theorem 1 (Cramér–Rao Lower Bound (CRLB))**.** *Let $\hat{\theta}$ be any unbiased estimator of θ_0 based on T i.i.d. observations $\{w_t\}_{t=1}^T$ with density $f(w \mid \theta)$ satisfying standard regularity conditions. Then*

$$\mathrm{Var}(\hat{\theta}) \succeq \frac{1}{T} I(\theta_0)^{-1},$$

where $\succeq$ denotes the positive semidefinite ordering.

Remark 2 (Interpretation)**.** The matrix $T^{-1}I(\theta_0)^{-1}$ is a lower bound on the covariance matrix of any unbiased estimator. An estimator that achieves this bound is said to be *(finite-sample) efficient.*

Definition 6 (Asymptotic Efficiency)**.** An estimator $\hat{\theta}_T$ is *asymptotically efficient* if

$$\sqrt{T}(\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}\big(0, I(\theta_0)^{-1}\big)\,.$$

Remark 3 (Relation to MLE)**.** Under correct specification and regularity conditions, the MLE is asymptotically efficient. That is, its asymptotic variance attains the inverse Fisher information.

3 Linear Regression Model (Running Example)

For $t = 1, \ldots, T$, observe (y_t, x_t) with $y_t \in \mathbb{R}$ and $x_t \in \mathbb{R}^k$. Let

$$y_t = x_t'\beta_0 + \varepsilon_t,$$

where $\beta_0 \in \mathbb{R}^k$ is the target parameter and ε_t is an error term. Stack $y \in \mathbb{R}^T$, $X \in \mathbb{R}^{T\times k}$ with rows x_t', and $\varepsilon \in \mathbb{R}^T$:

$$y = X\beta_0 + \varepsilon.$$

3.1 Classical assumptions

(A1) **Linearity:**

$$y = X\beta_0 + \varepsilon.$$

(A2) **Exogeneity.** One of the following is imposed, depending on context:

(A2a) **Strict Exogeneity:**

$$\mathbb{E}[\varepsilon_t \mid X] = 0 \quad \text{for all } t,$$

i.e. the error is mean independent of the entire regressor matrix (past, present, and future values).

(A2b) **Weak Exogeneity:**

$$\mathbb{E}[\varepsilon_t \mid x_t] = 0 \quad \text{for all } t,$$

i.e. the error is mean independent of the contemporaneous regressor only.

(A3) **Spherical Errors:**

$$\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_T$$

(homoskedastic, no serial correlation).

(A4) **Full Rank:**

$$X'X \text{ is invertible}$$

(no perfect multicollinearity).

(A5) **Normality (optional):**

$$\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_T).$$

Remark 4 (Strict vs Weak Exogeneity)**.** Strict exogeneity implies weak exogeneity, but not vice versa. In particular,

$$\mathbb{E}[\varepsilon_t \mid X] = 0 \;\Rightarrow\; \mathbb{E}[\varepsilon_t \mid x_t] = 0,$$

but $\mathbb{E}[\varepsilon_t \mid x_t] = 0$ does *not* rule out correlation between ε_t and past or future regressors. Strict exogeneity is typically required for unbiasedness of OLS in dynamic models and panel data settings whereas weak exogeneity is sufficient for consistency of OLS in many static cross-sectional and time-series regressions, but not for finite-sample unbiasedness in general.

Remark 5 (Orthogonality)**.** Under strict exogeneity,

$$\mathbb{E}[X'\varepsilon] = 0,$$

and under weak exogeneity,

$$\mathbb{E}[x_t \varepsilon_t] = 0 \quad \text{for each } t.$$

These orthogonality conditions motivate the moment restrictions used in GMM and IV estimation. They are *implied* by exogeneity assumptions and should not be interpreted as logically equivalent definitions.

Remark 6 (Implications)**.** Under (A1), (A2a), (A3), and (A4), OLS is BLUE by the Gauss–Markov theorem. Under (A1), (A2a), (A3), (A4), and (A5), OLS coincides with the Gaussian MLE for β, and exact finite-sample t and F inference (conditional on X) is available using estimators of σ^2.

4 OLS

4.1 Definition

OLS chooses β to minimize the sum of squared residuals:

$$\hat{\beta}_{\mathrm{OLS}} \in \arg\min_{\beta \in \mathbb{R}^k} (y - X\beta)'(y - X\beta).$$

Define the scalar objective function

$$Q(\beta) \equiv (y - X\beta)'(y - X\beta).$$

Expand:

$$Q(\beta) = y'y - 2\beta'X'y + \beta'X'X\beta.$$

The gradient of $Q(\beta)$ with respect to the vector β is

$$\nabla_\beta Q(\beta) = -2X'y + 2X'X\beta.$$

The first-order condition for a minimum is

$$\nabla_\beta Q(\beta) = 0 \quad \Longleftrightarrow \quad -2X'y + 2X'X\beta = 0 \quad \Longleftrightarrow \quad X'X\beta = X'y.$$

Under (A4) ($X'X$ invertible), the unique solution is

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y.$$

The corresponding normal equations are

$$X'(y - X\hat{\beta}_{\text{OLS}}) = 0.$$

Differentiate the gradient once more with respect to β:

$$\nabla^2_{\beta\beta} Q(\beta) = \frac{\partial}{\partial \beta'}\left(-2X'y + 2X'X\beta\right) = 2X'X.$$

Under (A4), $X'X$ is positive definite, hence

$$\nabla^2_{\beta\beta} Q(\beta) \succ 0.$$

Therefore $Q(\beta)$ is strictly convex in β, and the critical point

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y$$

is the *unique global minimizer* of the least squares objective.

4.2 Conditional unbiasedness under exogeneity

Assume (A1), (A2), and (A4). Using $y = X\beta_0 + \varepsilon$,

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'(X\beta_0 + \varepsilon) = \beta_0 + (X'X)^{-1}X'\varepsilon.$$

Taking conditional expectation given X:

$$\mathbb{E}[\hat{\beta}_{\text{OLS}} \mid X] = \beta_0 + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X].$$

Under (A2), $\mathbb{E}[\varepsilon \mid X] = 0$, hence

$$\mathbb{E}[\hat{\beta}_{\text{OLS}} \mid X] = \beta_0.$$

Thus OLS is conditionally unbiased.

4.3 Conditional variance under spherical errors

Assume (A1)–(A4). From $\hat{\beta}_{\text{OLS}} - \beta_0 = (X'X)^{-1}X'\varepsilon$,

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid X) = (X'X)^{-1}X'\text{Var}(\varepsilon \mid X)X(X'X)^{-1}.$$

Under (A3), $\text{Var}(\varepsilon \mid X) = \sigma^2 I_T$, so

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid X) = (X'X)^{-1}X'(\sigma^2 I_T)X(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

4.4 Gauss–Markov theorem (BLUE)

Theorem 2 (Gauss–Markov)**.** *Under (A1)–(A4), $\hat{\beta}_{OLS}$ is BLUE: among all estimators linear in y of the form $\tilde{\beta} = Ay$ that are unbiased conditional on X, OLS has the smallest conditional variance matrix in the positive semidefinite (PSD) ordering:*

$$\mathrm{Var}(\tilde{\beta} \mid X) - \mathrm{Var}(\hat{\beta}_{OLS} \mid X) \text{ is PSD.}$$

Remark 7 (Scope of Gauss–Markov)**.** Gauss–Markov does not require normality and does not claim optimality among *all* unbiased estimators. It is a comparison within the class of *linear* unbiased estimators under spherical errors.

5 MLE

5.1 Definition

Given a parametric density $f(w_t \mid \theta)$ (or $f(y \mid X, \theta)$), the MLE is any maximizer

$$\hat{\theta}_{\mathrm{MLE}} \in \arg\max_{\theta \in \Theta} \sum_{t=1}^{T} \log f(w_t \mid \theta).$$

5.2 Gaussian Linear Regression: MLE equals OLS for β

Assume (A1)–(A5): $y \mid X \sim \mathcal{N}(X\beta, \sigma^2 I_T)$. The log-likelihood (up to an additive constant) is

$$\ell(\beta, \sigma^2) = -\frac{T}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta).$$

Define

$$Q(\beta) \equiv (y - X\beta)'(y - X\beta).$$

Then

$$\nabla_\beta Q(\beta) = -2X'(y - X\beta).$$

Using the chain rule,

$$\nabla_\beta \ell(\beta, \sigma^2) = -\frac{1}{2\sigma^2}\nabla_\beta Q(\beta) = \frac{1}{\sigma^2}X'(y - X\beta).$$

FOC for β. The score equation for β is

$$\nabla_\beta \ell(\beta, \sigma^2) = 0 \quad \Longleftrightarrow \quad \frac{1}{\sigma^2}X'(y - X\beta) = 0 \quad \Longleftrightarrow \quad X'X\beta = X'y.$$

Under (A4) ($X'X$ invertible),

$$\hat{\beta}_{\mathrm{MLE}} = (X'X)^{-1}X'y = \hat{\beta}_{\mathrm{OLS}}.$$

Hessian. Differentiate the gradient again:

$$\nabla^2_{\beta\beta}\ell(\beta, \sigma^2) = \frac{\partial}{\partial \beta'}\left(\frac{1}{\sigma^2}X'(y - X\beta)\right) = -\frac{1}{\sigma^2}X'X.$$

Under (A4), $X'X$ is positive definite and $\sigma^2 > 0$, hence

$$\nabla^2_{\beta\beta}\ell(\beta, \sigma^2) \prec 0.$$

Therefore the critical point $\hat{\beta}_{\mathrm{MLE}}$ is a *strict global maximizer* of the log-likelihood in β.

FOC for σ^2. For completeness, the derivative with respect to σ^2 is

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{T}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{1}{(\sigma^2)^2}(y - X\beta)'(y - X\beta),$$

and setting this to zero at $\beta = \hat{\beta}$ yields

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{T}(y - X\hat{\beta})'(y - X\hat{\beta}).$$

5.3 Efficiency

Assume (A1)–(A5): $y \mid X \sim \mathcal{N}(X\beta_0, \sigma^2 I_T)$, with fixed X (or conditioning on X). The MLE for β attains the Cramér–Rao lower bound. The Fisher information matrix for β is

$$I_\beta(\beta_0) = \mathbb{E}\big[s_\beta(\beta_0, \sigma^2) s_\beta(\beta_0, \sigma^2)' \mid X\big].$$

Since $y - X\beta_0 = \varepsilon$ and $\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_T$,

$$s_\beta(\beta_0, \sigma^2) = \frac{1}{\sigma^2} X'\varepsilon,$$

so

$$I_\beta(\beta_0) = \frac{1}{\sigma^4} X'\mathbb{E}[\varepsilon\varepsilon' \mid X]X = \frac{1}{\sigma^4} X'(\sigma^2 I_T)X = \frac{1}{\sigma^2} X'X.$$

The CRLB for any unbiased estimator of β_0 (conditional on X) is

$$\text{Var}(\hat{\beta} \mid X) \succeq I_\beta(\beta_0)^{-1} = \sigma^2 (X'X)^{-1}.$$

From Section 4,

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid X) = \sigma^2 (X'X)^{-1}.$$

Therefore,

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid X) = I_\beta(\beta_0)^{-1},$$

so the MLE *attains the Cramér–Rao lower bound*. Hence, in the classical linear Gaussian model, OLS is not only BLUE but also *fully efficient* among all unbiased estimators.

6 GMM

6.1 Definition

Let $g(w_t, \theta) \in \mathbb{R}^m$ be a vector of moment functions. Suppose the model implies the population moment conditions

$$\mathbb{E}[g(w_t, \theta_0)] = 0.$$

Define the sample moment

$$\bar{g}_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} g(w_t, \theta).$$

Given a symmetric positive definite weighting matrix W_T, the GMM estimator is

$$\hat{\theta}_{\text{GMM}} \in \arg\min_{\theta \in \Theta} \ \bar{g}_T(\theta)' W_T \bar{g}_T(\theta).$$

6.2 Exact Identification vs Overidentification

If $m = k$ (number of moments equals number of parameters), the model is *exactly identified*. If $m > k$, it is *overidentified*. In the exactly identified case, if $\bar{g}_T(\theta) = 0$ has a (locally) unique solution, then that solution minimizes the GMM objective for any positive definite W_T. In the over identified case, W_T affects the point estimate and finite-sample behavior.

6.3 OLS as (exact-ID) GMM in Linear Regression

Under (A2), $\mathbb{E}[\varepsilon \mid X] = 0$ implies $\mathbb{E}[x_t \varepsilon_t] = 0$, hence the orthogonality condition

$$\mathbb{E}[x_t(y_t - x_t'\beta_0)] = 0.$$

Take

$$g(w_t, \beta) = x_t(y_t - x_t'\beta) \in \mathbb{R}^k.$$

Then

$$\bar{g}_T(\beta) = \frac{1}{T}\sum_{t=1}^{T} x_t(y_t - x_t'\beta) = \frac{1}{T}X'(y - X\beta).$$

In the exactly identified case, setting $\bar{g}_T(\beta) = 0$ yields

$$X'(y - X\beta) = 0 \implies \hat{\beta} = (X'X)^{-1}X'y = \hat{\beta}_{\text{OLS}}.$$

Thus OLS can be seen as exact-ID GMM for the moment condition $\mathbb{E}[x_t(y_t - x_t'\beta_0)] = 0$.

7 When Do OLS, MLE, and GMM Coincide?

In the linear regression model:

- OLS equals exact-ID GMM when the moments are $g_t(\beta) = x_t(y_t - x_t'\beta)$ and $X'X$ is invertible (so that $\bar{g}_T(\beta) = 0$ has a unique solution).
- OLS equals Gaussian MLE when, in addition, $y \mid X \sim \mathcal{N}(X\beta, \sigma^2 I_T)$.

Under (A1)–(A5), $\hat{\beta}_{\text{OLS}} = \hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{GMM}}$, where GMM uses the moment condition $g_t(\beta) = x_t(y_t - x_t'\beta)$. The equality is exact in finite samples, not merely asymptotic.

Remark 8. Use Cases

- **OLS** is simple, easily interpreted, and closed-form.
- **MLE** provides a full probabilistic model, useful for likelihood ratio tests, AIC/BIC, and extensions to nonlinear settings.
- **GMM** provides a framework driven by moment restrictions, extending naturally to endogeneity (IV), heteroskedasticity, dependence, and theory-based restrictions (e.g., Euler equations).

Remark 9 (Point Estimates vs Inference)**.** Even if $\hat{\beta}$ coincides, reported standard errors (and hence t-tests and confidence intervals) can differ depending on how one estimates $\text{Var}(\hat{\beta})$ and which assumptions are invoked (classical vs robust vs HAC vs bootstrap).

Remark 10 (Gauss–Markov)**.** Gauss–Markov is a benchmark optimality statement for OLS *under spherical errors* and *within linear unbiased estimators*. It does not address nonlinear estimators, misspecification, non-spherical errors (heteroskedasticity, serial correlation), endogeneity, or robust inference.

8 Breaking Classical Assumptions

This section breaks the classical assumptions one by one and considers implications for point estimates, estimator properties, and inference.

8.1 Non-normal Errors

Assume (A1)–(A4) hold, but drop normality (A5). Errors are non-normal, but remain spherical and exogeneity holds. Then OLS remains BLUE by Gauss–Markov, and remains conditionally unbiased (by (A2)) and consistent (under standard regularity). MLE depends on the likelihood assumed. If one *maintains* a Gaussian likelihood even though the true errors are non-normal, the resulting estimator for β still coincides with OLS (because the Gaussian score condition is $X'(y - X\beta) = 0$), and is interpreted as a *quasi*-MLE (QMLE).

Without normality, exact finite-sample t and F distributions are generally invalid. However, under standard conditions (e.g., i.i.d. with finite second moments), asymptotic normality of OLS can justify large-sample t-tests.

8.2 Heteroskedasticity

Assume (A1), (A2), (A4) hold, but replace (A3) with

$$\mathrm{Var}(\varepsilon \mid X) = \Omega \neq \sigma^2 I_T,$$

where Ω is diagonal but not proportional to identity (heteroskedasticity). Exogeneity holds. OLS still solves $X'(y - X\beta) = 0$, so the OLS point estimate does not change. Under exogeneity, OLS remains conditionally unbiased and consistent. However, Gauss–Markov does not apply, so OLS is generally not BLUE. If Ω were known, GLS would be more efficient. If one incorrectly maintains spherical Gaussian errors in likelihood-based inference, the resulting (classical) standard errors are generally inconsistent, and if one correctly models heteroskedasticity in the likelihood, the MLE generally changes and can be more efficient.

Remark 11 (Robust Variance: White.)**.** A heteroskedasticity-robust variance estimator is

$$\widehat{\mathrm{Var}}(\hat{\beta}_{\mathrm{OLS}})_{\mathrm{robust}} = (X'X)^{-1} \left(\sum_{t=1}^{T} x_t x_t' \hat{\varepsilon}_t^2 \right) (X'X)^{-1}.$$

Remark 12 (Serial Correlation)**.** Assume $\mathbb{E}[\varepsilon \mid X] = 0$ but allow $\mathrm{Cov}(\varepsilon_t, \varepsilon_s \mid X) \neq 0$ for $t \neq s$. Exogeneity holds, so the OLS point estimate again does not change, but standard errors must account for dependence (e.g., HAC/Newey–West).

8.3 Endogeneity

Break exogeneity:

$$\mathbb{E}[\varepsilon \mid X] \neq 0 \quad \text{(equivalently, conditional mean zero fails).}$$

Then OLS is generally biased (finite sample) and inconsistent. MLE is also inconsistent unless the likelihood correctly models the endogeneity (e.g., joint modeling, control functions).

IV/GMM. If valid instruments z_t exist such that $\mathbb{E}[z_t\varepsilon_t] = 0$ and the population rank condition holds,

$$\operatorname{rank}\big(\mathbb{E}[z_t x_t']\big) = k,$$

then β_0 is identified by the moment conditions

$$\mathbb{E}[z_t(y_t - x_t'\beta_0)] = 0.$$

Let $z_t \in \mathbb{R}^m$ with $m \geq k$ and define

$$g_t(\beta) = z_t(y_t - x_t'\beta) \in \mathbb{R}^m.$$

In exact identification ($m = k$) and with Z denoting the $T \times k$ matrix with rows z_t', the sample moment condition is

$$\sum_{t=1}^{T} z_t(y_t - x_t'\beta) = 0 \quad \Longleftrightarrow \quad Z'(y - X\beta) = 0,$$

which yields an IV-type solution (when the relevant matrices are invertible).

8.4 Overidentification and the J-test

If $m > k$, the point estimate depends on W_T (and typically on how W_T is estimated). Overidentifying restrictions can be tested using the J-statistic. When $\hat{\theta}$ is the efficient two-step GMM estimator and $\widehat{W}$ consistently estimates the optimal weight matrix (typically the inverse of the long-run covariance of $\sqrt{T}\bar{g}_T(\theta_0)$),

$$J_T = T \cdot \bar{g}_T(\hat{\theta})'\widehat{W}\,\bar{g}_T(\hat{\theta}) \;\Rightarrow\; \chi^2_{m-k}$$

under correct specification (with regularity). Although additional valid moments can reduce asymptotic variance under optimal weighting, in finite samples weak or noisy moments can worsen mean squared error.

9 Instrumental Variables and GMM

The relationship between Instrumental Variables (IV), Two-Stage Least Squares (2SLS), and GMM can be summarized as *2SLS is a special case of GMM*, and efficient GMM generalizes 2SLS when errors are heteroskedastic or serially correlated. This section provides the details.

9.1 Setup

Consider the linear model

$$y = X\beta_0 + \varepsilon,$$

where some columns of X are endogenous, i.e.

$$\mathbb{E}[X'\varepsilon] \neq 0.$$

Let $Z \in \mathbb{R}^{T\times m}$ be a matrix of instruments with rows z_t' satisfying:

(IV1) **Instrument Exogeneity:**

$$\mathbb{E}[z_t\varepsilon_t] = 0.$$

(IV2) **Instrument Relevance:**

$$\operatorname{rank}\big(\mathbb{E}[z_t x_t']\big) = k.$$

These conditions imply the population moment restrictions

$$\mathbb{E}[Z'(y - X\beta_0)] = 0.$$

9.2 IV as exact-ID or over-ID GMM

Define the moment function

$$g_t(\beta) = z_t(y_t - x_t'\beta) \in \mathbb{R}^m.$$

Then the sample moment is

$$\bar{g}_T(\beta) = \frac{1}{T}\sum_{t=1}^{T} z_t(y_t - x_t'\beta) = \frac{1}{T}Z'(y - X\beta).$$

The GMM estimator solves

$$\hat{\beta}_{\text{GMM}} \in \arg\min_{\beta}\ \bar{g}_T(\beta)' W_T \bar{g}_T(\beta),$$

where W_T is symmetric positive definite.

Exact identification ($m = k$). If the number of instruments equals the number of regressors, the model is exactly identified and the moment condition

$$Z'(y - X\beta) = 0$$

pins down β (locally). Solving,

$$Z'X\beta = Z'y \quad \Longrightarrow \quad \hat{\beta} = (Z'X)^{-1}Z'y.$$

This is the just-identified IV estimator.

Overidentification ($m > k$). If $m > k$, the model is overidentified and no β satisfies all moments exactly. GMM chooses $\hat{\beta}$ to make the moments "as small as possible" in a quadratic sense.

9.3 Deriving 2SLS from GMM

The classical 2SLS estimator is

$$\hat{\beta}_{\text{2SLS}} = (X'P_Z X)^{-1}X'P_Z y, \quad P_Z \equiv Z(Z'Z)^{-1}Z'.$$

Note that this is a GMM estimator. Let

$$W_T = (Z'Z/T)^{-1}.$$

Using $\bar{g}_T(\beta) = T^{-1}Z'(y - X\beta)$,

$$Q_T(\beta) = \bar{g}_T(\beta)' W_T \bar{g}_T(\beta) = (y - X\beta)'Z(Z'Z)^{-1}Z'(y - X\beta) = (y - X\beta)'P_Z(y - X\beta).$$

Define

$$Q_T(\beta) = (y - X\beta)' P_Z (y - X\beta).$$

Expand:

$$Q_T(\beta) = y' P_Z y - 2\beta' X' P_Z y + \beta' X' P_Z X \beta.$$

Take the gradient:

$$\nabla_\beta Q_T(\beta) = -2X' P_Z y + 2X' P_Z X \beta.$$

Set equal to zero:

$$-2X' P_Z y + 2X' P_Z X \beta = 0 \quad \Longleftrightarrow \quad X' P_Z X \beta = X' P_Z y.$$

Therefore,

$$\hat{\beta} = (X' P_Z X)^{-1} X' P_Z y = \hat{\beta}_{\text{2SLS}}.$$

Thus the 2SLS is the GMM estimator with $g_t(\beta) = z_t(y_t - x_t'\beta)$ and $W_T = (Z'Z/T)^{-1}$.

9.4 Efficient GMM vs 2SLS

The choice $W_T = (Z'Z/T)^{-1}$ is *not* generally optimal. The optimal GMM weighting matrix is

$$W^* = S^{-1}, \quad S \equiv \operatorname{Var}\Big(\sqrt{T}\, \bar{g}_T(\beta_0)\Big) = \mathbb{E}[z_t z_t' \varepsilon_t^2].$$

- If $\operatorname{Var}(\varepsilon_t \mid Z) = \sigma^2$ (homoskedasticity), then

  $$S = \sigma^2 \mathbb{E}[z_t z_t'],$$

  and efficient GMM coincides with 2SLS (up to scale).

- If $\operatorname{Var}(\varepsilon_t \mid Z)$ is heteroskedastic, then

  $$S \neq \sigma^2 \mathbb{E}[z_t z_t'],$$

  and efficient GMM differs from 2SLS. In this case, 2SLS is consistent but not efficient, and GMM with $W^* = S^{-1}$ improves efficiency.

10 MLE as GMM on the Score

MLE can be thought of as GMM on the score. Let $\ell_t(\theta) = \log f(w_t \mid \theta)$ and define the score

$$s_t(\theta) = \frac{\partial \ell_t(\theta)}{\partial \theta}.$$

Note that $\mathbb{E}[s_t(\theta_0)] = 0$ under correct specification. Assume $f(\cdot \mid \theta)$ is a correctly specified density and regularity conditions permit differentiation under the integral sign. Then

$$\begin{aligned}
\mathbb{E}[s_t(\theta_0)] &= \int \frac{\partial}{\partial \theta} \log f(w \mid \theta)\Big|_{\theta=\theta_0} f(w \mid \theta_0)\, dw \\
&= \int \frac{\partial}{\partial \theta} f(w \mid \theta)\Big|_{\theta=\theta_0} dw \\
&= \frac{\partial}{\partial \theta} \int f(w \mid \theta)\, dw\Big|_{\theta=\theta_0} \\
&= \frac{\partial}{\partial \theta} 1 \\
&= 0.
\end{aligned}$$

Thus $s_t(\theta)$ defines valid moment conditions. Under exact identification and regularity, consider GMM with moments *equal to the score*:

$$g_t(\theta) = s_t(\theta), \qquad \bar{s}_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} s_t(\theta).$$

The GMM objective is

$$Q_T(\theta) = \bar{s}_T(\theta)' W_T \bar{s}_T(\theta),$$

with W_T symmetric positive definite. If the score is k-dimensional for k parameters (exact identification) and there is a unique interior maximizer of the likelihood, then the MLE satisfies the sample first-order conditions

$$\bar{s}_T(\hat{\theta}_{\text{MLE}}) = 0.$$

Any θ that solves $\bar{s}_T(\theta) = 0$ achieves $Q_T(\theta) = 0$, the minimum possible value. Under the uniqueness/interiority condition ensuring the score root corresponds to the likelihood maximizer,

$$\hat{\theta}_{\text{MLE}} = \hat{\theta}_{\text{GMM(score)}}.$$

As an example, in Gaussian linear regression, the score for β is

$$s_\beta(\beta, \sigma^2) = \frac{1}{\sigma^2} X'(y - X\beta).$$

Setting $s_\beta = 0$ yields $X'(y - X\beta) = 0$, hence $\hat{\beta} = (X'X)^{-1}X'y$.

11 Summary

The table below summarizes key results.

Method	**Input/Assumption**	**Objective**	**Use Cases**
OLS	Linear conditional mean, exogeneity (for unbiasedness)	Minimize squared residuals (projection)	Baseline linear regression; transparent reduced-form estimation
MLE	Full parametric distribution specified	Maximize likelihood (density fit)	Discrete choice, nonlinear models, state-space, regime switching; likelihood-based tests
GMM	Moment restrictions (orthogonality), possibly instruments	Match moments via quadratic criterion	IV/endogeneity; theory-based moments (Euler equations); robust estimation under weak distributional assumptions

Table 1: High-level comparison of OLS, MLE, and GMM.